

Loops (break/continue)

break & continue

Two important concepts one should learn with loop are **break** & **continue**. These two keywords grant us more control when running the loop.

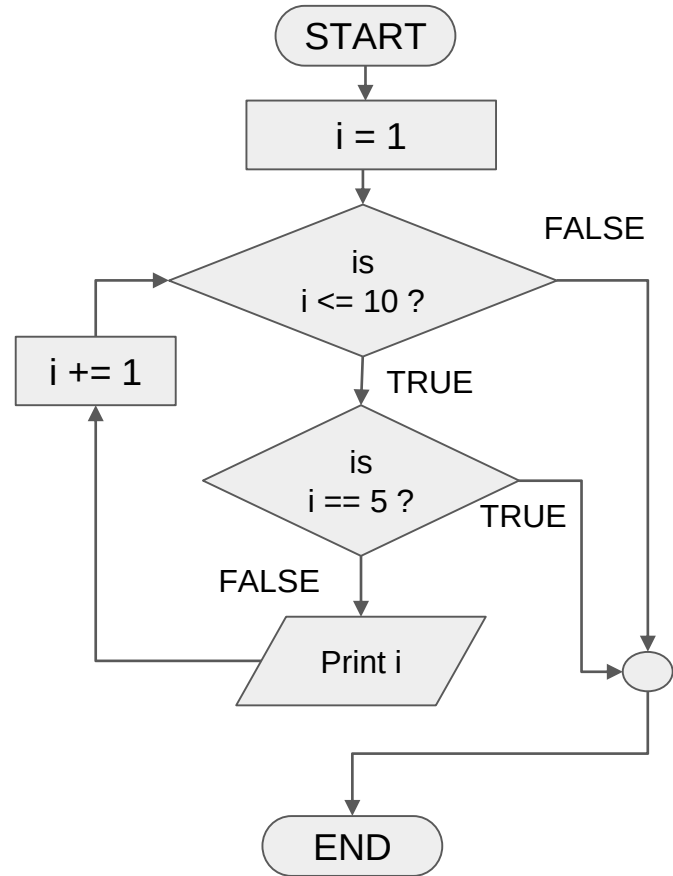
- ❑ **break**: the break keyword allows us to **stop the loop** early.
- ❑ **continue**: the continue keyword allows us to skip the rest of the loop body and start the next iteration of the loop.

Note: both break and continue are written inside the loop's body and are generally bounded by a condition.

break keyword - visualization

Scenario 4: Draw a flowchart that prints all numbers from 1 to 10 but when 5 is reached the program stops.

As you can see in the flow chart, when **i==5** is **true** we're breaking out of the loop and directly reaching the end.



break keyword - code

Comparison of Scenario 4 flowchart and Java code using for loop.

```
1. public class BreakScenario4{
2.     public static void main(String[] args) {
3.         for (int i=1 ; i<=10 ; i++) {
4.             if ( i == 5 ) {
5.                 break;
6.             }
7.             System.out.print( i + " ");
8.         }
9.     }
10. }
```

Welcome to DrJava. Working directory is /home/Java110

> run BreakScenario4

1 2 3 4 >

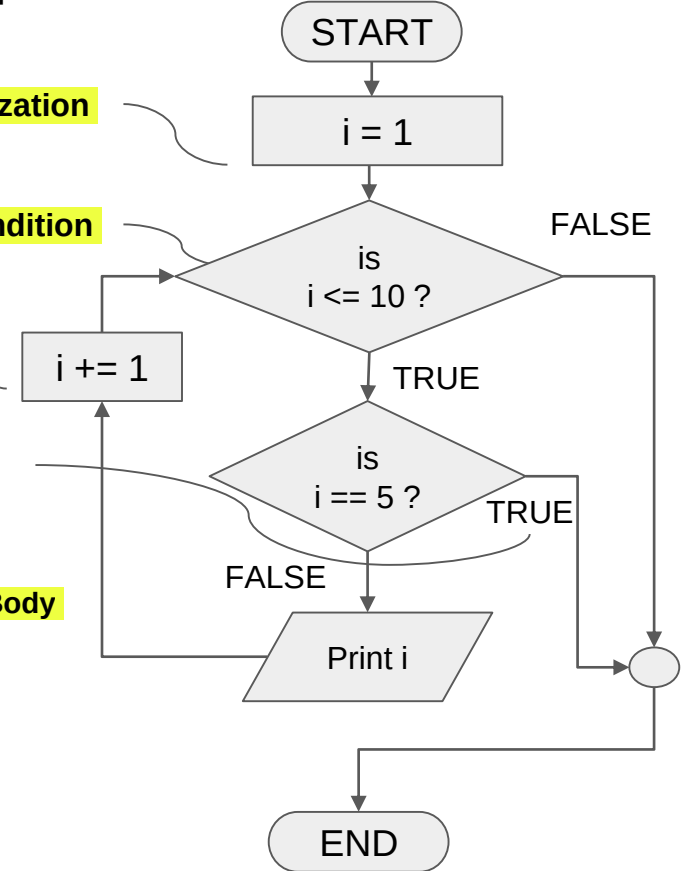
Initialization

Condition

Steps

Break

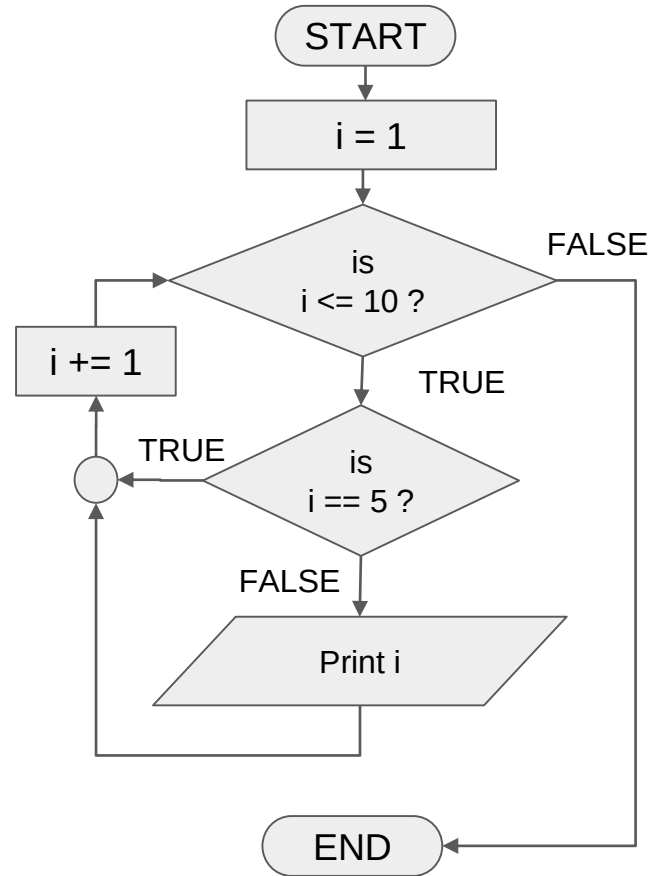
Body



continue keyword - visualization

Scenario 4: Draw a flowchart that prints all numbers from 1 to 10 except for 5.

As you can see in the flow chart, whenever **i==5** is **true** we're skipping the next line and going to the next iteration.



continue keyword - code

Comparison of Scenario 4 flowchart and Java code using for loop.

```
1. public class ContinueScenario4{
2.     public static void main(String[] args) {
3.         for (int i=1 ; i<=10 ; i++) {
4.             if ( i == 5 ) {
5.                 continue;
6.             }
7.             System.out.print( i + " ");
8.         }
9.     }
10. }
```

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> run ContinueScenario4

1 2 3 4 6 7 8 9 10 >

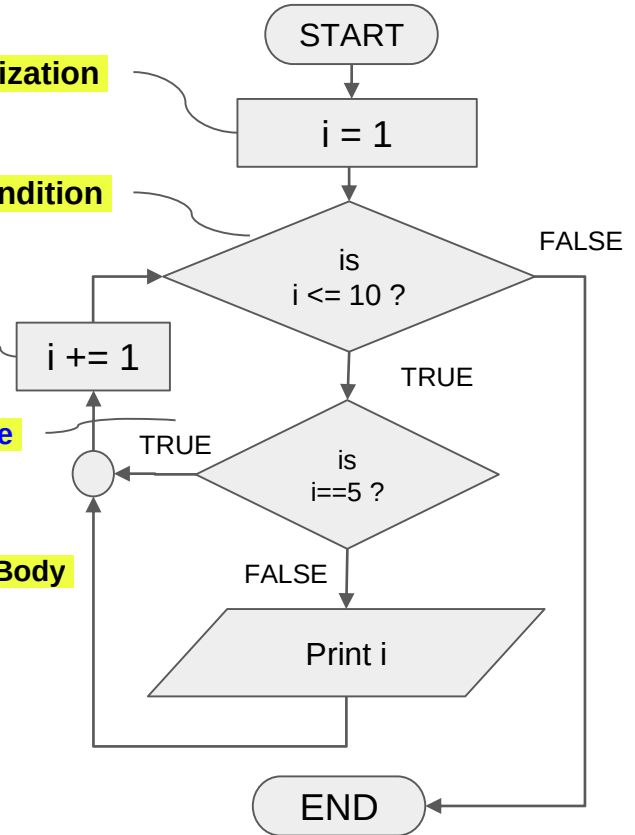
Initialization

Condition

Steps

Continue

Body



continue keyword - code

This was an example of **continue** keyword using **for** loop. Now, what would happen if instead of **for** loop you used **while** loop?

That's for you to find out!

THANK YOU!